

One construction, chosen coefficients

Every vector operation is $c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k$ with the coefficients fixed in advance. What separates the entries below is only which coefficients are allowed — pinned to specific values, or left free.

COEFFICIENTS PINNED — THE BASIC OPERATIONS

4

1	Scalar multiplication one vector, one coefficient	$c\mathbf{v}$ The shortest combination there is. Its span is the line through $\mathbf{v}$, which is the $k = 1$ case of everything below — see basic operations .
2	Addition both coefficients 1	$1 \cdot \mathbf{u} + 1 \cdot \mathbf{v}$ Addition is not a separate operation that combinations generalise — it is the combination with both coefficients set to one. The tip-to-tail picture is the geometric reading of exactly this.
3	Subtraction coefficients 1 and -1	$1 \cdot \mathbf{u} + (-1) \cdot \mathbf{v}$ Likewise, with one coefficient negated. This is why subtraction needs no rules of its own: everything true of combinations applies to it directly.
4	Trivial combination always available, always gives $\mathbf{0}$	$0\mathbf{v}_1 + \cdots + 0\mathbf{v}_k = \mathbf{0}$ Available for any set whatsoever, which is why finding a combination equal to $\mathbf{0}$ proves nothing. Independence asks whether it is the only one — the whole subject turns on that word.

COEFFICIENTS FREE — WHAT THE CONSTRUCTION GENERATES

3

5	General combination arbitrary scalars	$c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k$ The definition the rest are instances of. Note it is a finite sum with arbitrary real coefficients — no convergence question arises, which is what separates this from an infinite series.
6	Span the set of every reachable vector	$\text{Span}\{\mathbf{v}\} = \{\text{all combinations}\}$ Let the coefficients range over everything and collect the results. The span is automatically a subspace — a combination of combinations is a combination — which is the cheapest subspace proof available.
7	Membership as a system columns of A are the $\mathbf{v}_i$	$\mathbf{b} \in \text{Span} \iff A\mathbf{c} = \mathbf{b} \text{ solvable}$ The bridge to computation. Asking whether $\mathbf{b}$ is a combination is asking whether a linear system is consistent, and the coefficients are its solution — which is why row reduction answers questions about spans.

The progression is one of freedom. Fix every coefficient and you get a single operation; free them and you get the span; ask whether a particular combination can reach $\mathbf{0}$ and you get independence. Nothing new is introduced at any stage — only the question asked about the same expression.